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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
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B.Tech. 3nd Year/ 6th Semester

Assignment -4

Subject Name: Data Science
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Q1) Explain the concept of data wrangling and its importance in data analysis.

Ans) Data wrangling is the process of cleaning, transforming and organizing raw data into a usable format for analysis.

Step =

- Data Collection
- Data Cleaning
- Data transformation
- Data Integration
- Data Reduction

* Importance = Improves data quality, makes analysis accurate and reliable, saves time during analysis, helps in decision making

Q2) explain string manipulation in python with suitable code examples

Ans String manipulation means modifying and working with text data using python

① Changing Case

```
text = "hello world"
print (text.upper())
print (text.lower())
```

② Removing Spaces

```
text = " Hello"
print (text.strip())
```

③ Replacing Text

```
text = "I like Java"
print (text.replace ("Java", "Python"))
```

④ Splitting Strings

```
text = "apple, banana, mango"
fruits = text.split (",")
print (fruits)
```

5) Joining strings

```
words = ["Hello", "World"]
print(" ".join(words))
```

6) finding substrings

```
text = "Python Programming"
print(text.find("Pro"))
```

Q3) Explain the general techniques and programming practices used for efficient handling of large datasets in data science.

Ans) Handling large datasets efficiently is important in data science to improve performance and reduce memory usage.

Key techniques:

1) Use efficient libraries

- Use Pandas, numpy for optimized operations.

2) Chunk processing

Process data in smaller parts instead of loading all at once

```
import pandas as pd
for chunk in pd.read_csv("file.csv", chunksize = 1000):
    process(chunk)
```

③ Use vectorization

- Avoid loops, use built-in functions

df["new"] = df["col"] * 2

④ Memory optimization

- Use correct data types

df["age"] = df["age"].astype("int8")

⑤ Use databases

→ Store large data in SQL instead of memory

⑥ Parallel processing

→ Use libraries like multiprocessing or Dask

⑦ Data sampling

→ Use smaller samples for testing

(Q4) Explain different data transformation techniques used in preprocessing.

Ans Data transformation converts raw data into a suitable format for analysis or modeling.

1) Normalization

Scale data below 0 & 1.

x_norm = (x - min(x)) / (max(x) - min(x))

2) Standardization

convert data to mean = 0 & std = 1

x_std = (x - mean(x)) / std(x)

3) Encoding Categorical Data

. convert text into no

pd.get_dummies(df['Color'])

4) Handling missing values

→ fill or remove data

→ df.fillna(0)

5) Feature Scaling

. assume all features have equal importance

6) log transform

. Reduce skewness

import numpy as np

df['log_col'] = np.log(df['col'])